

Department of Computer Science

Nnamdi Azikiwe University, Awka

Course Content Specifications/Syllabus of all Courses

CSC100 COMPUTER APPRECIATION (3 UNITS)

History of computer; the functional components of a digital computer, characteristics of a computer. The use of computer in several spheres of life;- business, commerce, planning (quantitative and qualitative), jurisprudence, education, Medicare, Data Communication and Networks, Internet, Economy, Politics/governance, etc. Computer Operations: Introduction to DOS and WINDOWS environment; Treatment of some specific and common application packages for data processing, such as word processor, spreadsheet, desktop publishing packages, etc.

CSC101: INTRODUCTION TO COMPUTER PROGRAMING (3 UNITS)

Introduction to Computer Systems - History and Development; Computer development timeline, functional unit of the Computer, primary and secondary memories, core memory etc. Magnetic devices: disks, tapes, video disks, flash drives, memory cards. Peripheral devices, printers, monitors, data entry and pointing devices, touch screens, biometric devices-fingerprint etc. characteristics, applications/uses, The number system; algorithm and flowchart; introduces students to visual IDE's e.g. VB.NET, visual C#.NET, Visual C++.NET; starting, environment (tool bars, toolbox, properties, explorer etc.) creating and saving a project; solving simple programming tasks, programming concepts: symbolic names, lists, arrays, expressions, control statements and functions.

CSC102: INTRODUCTION TO COMPUTER PROGRAMMING (3 UNITS)

Problem solving strategies; Role of algorithm in problem solving process, implementation strategies, concepts and properties of algorithms, UML diagrams, control structures, assignment statements, input/output, arrays and subscripts, array manipulations. Using Visual Basic to solve practical problems.

CSC112: COMPUTER APPLICATION AND MANAGEMENT (2 UNITS)

Pre-requisite CSC 101

Overview of hardware components ; using the computer: application areas, operating systems-(windows) and application packages, classes (word processors, spreadsheet, graphics etc.; MS word (creating documents, saving, editing, formatting and table), MS Access (creating a database, defining fields types and records, creating tables, queries and reports), MS- excel (worksheets, entering data, graphs, formula etc.), MS-PowerPoint (preparing presentations, slide transitions, inserting multimedia, rehearse time etc.); computer viruses: definition, causes, modes of transmission, preventive measures-antivirus kits, managing a computer centre, operating procedure, performance and security.

CSC201: COMPUTER PROGRAMMING I (2 UNITS)

Pre-requisite CSC 101 and CSC 102

Introduction to Computer Systems and programming C++ will be used in teaching the above C ++ background; development interface (creating documents, saving, editing, formatting and tables) object orientations: classes and methods; basic structure; input /output, keywords and operators, variables/constants, flow control-iteration, decision; data types and structure: numeric and string types, arrays, pointers; operations, arithmetic and string operations, data manipulation and file processing.

CSC211: INTRODUCTION TO DIGITAL DESIGN (2 UNITS)

Combination of Logic, sequential logic, microprocessor. Functional description ,microprocessor architecture, systems operation and interfacing, types on instructions and instruction formats, number systems and codes, binary coded decimals, Excess -3 code, gray codes; error detection and correction, parity method for error detection, logic gates, circuit symbols logic manipulations- Booleans Algebra, Karnaugh map (K-map) to

simplify logic circuits. Logical implementation using relays and gates. Using Quine McClusky to simplify logic circuits. A specific microcomputer based design may be used to teach

CSC221: ASSEMBLY LANGUAGE PROGRAMMING (2 UNITS)

Pre-requisite CSC 101, CSC 102, CSC 113

Introduction to Machine-level programming, addressing techniques, at least 16-bit instruction set and addressing modes, macros, assembler, segmentation and linkage; assembler construction, interpretation routines.

CSC231: AUTOMATA THEORY, COMPUTABILITY AND FORMAL LANGUAGE (2 UNITS)

Automata theory, formal grammar and automata; regular languages, context free languages, deterministic parsing of content-free language, recursive language.

CSC261: INFORMATION TECHNOLOGY (2 UNITS)

Overview of Computers, IT and business; hardware- physical components; software- system and application; connectivity-networks and work groups. Computer in society security, ethics and law; global community-internet, e-mail, social network groups. Challenges environment: recycling etc.; power: energy waste etc. health strains, radiation etc.

CSC202 COMPUTER PROGRAMMING 11 (3 UNITS)

Prerequisite: CSC201

Principles of good programming; objected orientation revisited; java IDE (creating, saving and opening of program files); classes and methods revisited; core concepts: input, output, variables/ constants, iteration, decision; data types and structure; numeric and string types, arrays pointers etc.; operations: arithmetic and string operations, data manipulation, advanced concepts; inheritance, polymorphism, encapsulation handling; java IO; file manipulation

CSC222: DATA MANAGEMENT (2 UNITS)

Prerequisite: CSC201

Data storage and retrieval, information: capture and representation, management applications, analysis and indexing search retrieval, privacy, integrity, security, scalability, efficiency and effectiveness; data types, records and files. Files processing methods, mapping logical organization on to physical storage. Backup procedures and file processing.

CSC232: DATABASE DESIGN AND MANAGEMENT (2 UNITS)

Introduction to database systems: Concept of relational database and design principles using entity relationship model. DBMS functions, database architecture and data independence, features and concepts field, tables, queries, forms, reports. Design and development of database applications using a current DBMS e.g. MS-Access, create, modify and query a database.

CSC252: DATA STRUCTURES (2 UNITS)

Pre-requisite CSC 201 and CSC 102

Primitive types, arrays, records, strings and string processing, data representation in memory, stack and heap allocation. Implementation strategies for queue, stacks, trees, runtime storage management: pointers and references, linked structures. Use C, C++ or java for implementation.

CSC311: DIGITAL DESIGN AND MICROPROCESSORS (2 UNITS)

Introduction to microprocessor architecture-systems operations and interfacing, Encoders, decoders and multiplexers. Registers and counters. Arithmetic circuits: binary adders-parallel and serial adders; full and half adders, binary subtractors - full and half subtractors, At least 16 or 32-bit microprocessor (Architecture and instruction set). Microcomputer interfacing, microcomputer-based A-D and D-A converters. Design, construction, programming and testing of microprocessors control systems.

CSC321: OPERATING SYSTEM (3 UNITS)

Prerequisite: CSC211, CSC261

Overview of operating systems: roles and purpose, basic functions device management; processes, threads and device management, memory management, file management, OS strategies and technologies, batch systems, time sharing systems mechanism to support client server models, hand held device, embedded systems, influences of network security and multimedia. OS principles: interrupts, concurrency, process scheduling strategies. Deadlock-prevention, avoidance, detection and recovery, manual deadlock management. Basic synchronization principles. Virtual memory-address translation, paging and segmentation.

CSC341: COMPUTATIONAL SCIENCE AND NUMERICAL METHOD (2 UNITS)

Prerequisite: CSC202, MAT202, MAT204

Application of mathematical packages in solving numerical problems such as floating point arithmetic, numerical integration and differentiation, solution of nonlinear equations, solution of ordinary differential equation (ODE), error analysis and norms, Eigen values and Eigen vectors, boundary value problems for ODEs, non-linear systems of algebraic equations. Laboratory course administered mainly with MATLAB.

CSC351: SYSTEM ANALYSIS AND DESIG (2 UNITS)

Overview of industry standard software engineering methodologies; hands on experience with case studies in developing systems from preliminary investigations to object oriented analysis and design; use of modeling tools, prototyping, presentations and technical reports. Implementation (includes verification, validation and test) and system maintenance issues. System development life cycle (SDLC), Rapid Application development (RAD), Agile Approach, Extreme programming and Rational Unified Process. CASE tools for automating the modeling of UML-based object-oriented models.

CSC361: OPERATION RESEARCH (2 UNITS)

The nature of Operation Research, allocation; inventory problems; replacement maintenance and reliability problems. Dynamic programming: sequencing and co-ordination.

CSC371: INFORMATION SYSTEMS AND MANAGEMENT (2 UNITS)

Information system concepts, approaches and methodologies, system design and implementation, architecture. Benefits and challenges, system performance and standards information systems development toolkit, MIS, management of organizations information systems and database theory and management, risks with MIS, Decision support system (DSS), Transaction processing systems. Experts system, communication and collaboration system prerequisites, system analysis and design, knowledge Asset management. Career development.

GIT399: INDUSTRIAL TRAINING (6 UNITS)

Require 6 months of industrial Training. Students experience will be documented and presented in a seminar.

CSC401: DATA COMMUNICATION/NETWORKS (2 UNITS)

Prerequisite: CSC211, CSC321

Introduction to the concepts and terminology of computer systems communications; network, LAN, WAN, internet; communication layers, OSI model, TCP/IP, network topologies, communication standards and protocols and IP addressing, network hardware and function. Distributed computing concepts and motivation, mobile/wireless computing, client/server and web application, cloud computing, communication, mechanism, synchronization, naming, replication and coherence, fault tolerance, security, access control, key management, cryptography, distributed file systems: NTFS, CODA etc.

CSC411: COMPILER CONSTRUCTION (2 UNITS)

Prerequisite: CSC231

Overview of compilers, assemblers and interpreters, structures and functional aspects of a typical compiler; syntax, semantics and pragmatics. Functional relationships between lexical analysis, expression and analysis and code generation. Grammars and languages, parsing, scanner, recognizers, runtime storage organization-top-down and bottom up, use of display allocation of storage to runtime variables.

CSC421: SURVEYING OF PROGRAMMING LANGUAGES (2 UNITS)

Overview of programming languages. Brief survey of programming paradigms (procedural, objected orient-oriented, functional, declarative non algorithm, scripting), the effects of scale on programming methodology. Types, declaration modes, (binding, visibility, scope and life time) type checking, garbage collection, abstract mechanism: procedures, functions and iterations and abstraction mechanism, parameterization mechanism (reference vs. value); activation records and storage management, object oriented language paradigms, functional and logic language paradigms.

CSC431: ALGORITHMS AND COMPLEXITY ANALYSIS (2 UNITS)

Principles of good programming style, expression, and documentation, structured programming concepts, debugging, testing, verifying, code inspection, semantic analysis, string processing, data structure. Recursion; efficiency of algorithm. Algorithm strategies: fundamental computing, numerical, sequential and binary search trees, hash table, graph and its representation. Basic algorithmic analysis.

CSC441: RESEARCH METHODS AND SEMINAR (2 UNITS)

General concept of research methodology, nature and objectives of scientific research, designing research proposals. Presentation and structure of research in Computer Science. Review of literature, analysis and methods, introduction to inquiry, structuring inquiry, research design, conceptualization and measurement, logic of sampling, modes of observation, field and lab research, content analysis and analysis of existing data, experiments, evaluation research, survey research, analysis of data, qualifying data, elementary analysis, index and scale construction. Properties of evaluation instrument, nature of data and reporting results

CSC451: ARTIFICIAL INTELLIGENCE (2 UNITS)**Prerequisite: CSC231**

Introduction: definitions, Turing tests, AI systems: language translation: application- Air traffic control, supervisory systems. Robots, Autonomous vehicles etc. knowledge representation-uncertain knowledge and reasoning, non-monotonic reasoning. Case based reasoning, cognitive modeling, distributed AI, planning, learning, problem solving and search, vision, automated reasoning , natural language processing , agents introduction of LISP, recursion, user defined functions, higher order functions, lambda, expression. LISP structure box and pointer notation dotted pairs, binary trees, input/output

CSC461: OBJECT ORIENTED PROGRAMMING (3 UNITS)

Introduction to object oriented concepts, classes and objects; encapsulation, inheritance , polymorphism and abstraction, static and virtual methods, class structure, object variables and the new operator, using objects, arrays of objects, methods and fields, using this reference, garbage collection and the finalize methods, static fields, static methods, parameter passing.

CSC471: WEB DESIGN, APPLICATION AND MANAGEMENT

Overview of the internet; origin, history and development, internet protocol-TCP/IP (Transfer Control protocol/ internet Protocol) intranet, extranet. Connecting to the internet: low and high bandwidth; E-communication, e-mail, instant messages, text messages, VoIP, online conferencing, chat rooms, social networking sites, blogs, message boards or newsgroups, understanding an e-mail address and interface. Components: address line, subject line, body and attachment, website design: HTML, PHP, JavaScript. Creating a simple web application: PHP and MySQL, practical approaches to managing complexity and improving interactivity

CSC412: COMPUTER GRAPHICS AND VISUALIZATION (2 UNITS)

Hardware components required for graphics, plotters, microfilm, graphic tablets, light pen, other graphical input aids. Refresh display, buggers, changing images, light pen interactions. 2 dimensional and 3 dimensional transformation perspective, clipping algorithm, hidden live removal, bolden surface removal: warlocks methods, shading, data reduction for graphical input. Introduction to handwriting and character recognition.

CSC422: SOFTWARE DEVELOPMENT AND MANAGEMENT (2 UNITS)

Software design: software architecture, design patterns, object oriented analysis and design, using APIs, class browser, component based programming, requirement analysis and design modeling tools. Testing tools, integration mechanism, automated processes. Standards: project control, project team, project implementation and documentation

CSC442: COMPUTER MODELLING & SIMULATION (2 UNITS)

Basic definition and uses, simulation process, some basic statistic distributions theory, model and simulation, queues, basic concepts, Kendal notation, queuing rules, Little's laws, queuing networks, types of queues. Birth-death processes, poison processes, random numbers and types, Exercises.

CSC452: EXPERT SYSTEMS (2 UNITS)**Prerequisite CSC351**

Introduction to knowledge engineering and expert systems, knowledge representation, inference methods, knowledge acquisition and explanation, reasoning under uncertainty, design of expert system. Objectives, knowledge, reasoning, ES life cycle, ES implementation Salience, rete algorithm, ES technology: tools, shells, languages. ES elements, facts, rules, inference mechanism, important concepts and terms. Semantic nets, frames, predicate logic. CLIPS overview.

CSC462: COMPUTER INSTALLATION AND MANAGEMENT (2 UNITS)

The concepts of computerization, role of computer centres, general operating procedures, network facilities and access control to resources. Hardware devices, assembly and installation, operating systems and network management, shared resources: printers, scanners and cameras, standard software installation and maintenance, purchase and renewal of software licenses. System routine maintenance and virus protection. Hardware and software updates and replacement plans.

CSC482: COMPUTER ARCHITECTURE (2UNITS)

Basic understanding of computer architecture, hardware and its functions, internal coding of data and instructions, logic circuits, microprocessor instruction executions, basic assembler programming with practical instruction and instruction formats. Data addressing methods and memory segmentations. Memory system addressing hierarchy, control systems-virtual memory, hardware.

CSC499: PROJECT (6 UNITS)

Each student should embark on work that will lead to substantial software development under the supervision of a member of staff.

CSC492: INTRODUCTION TO REMOTE SENSING AND GIS (ELECTIVE)(2 UNITS)

Principle of remote sensing, detection of objects, change detection, image analysis, signal processing, principles of global positioning system, map development, environmental analysis, features and topographic analysis, land use allocation. Any of the following software could be used, Archview, ArchGIS, Archinfo, Idiris or PCI image works



M.O. Onyesolu

<https://orcid.org/0000-0003-3357-4847>

<http://unizik.edu.ng/onyesolumo>